

RELTWIN: CONTRASTING LOCALLY VALID TIMESTAMP ANSWERS FOR QUERY-SPECIFIC AUDIO GROUNDING

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ABSTRACT

Audio temporal grounding locates the time intervals described by a text query. When the same sounds recur in different orders, the model must select the occurrence that matches the query. We introduce RelTwin, a fine-tuning framework that compares complete timestamp answers. Reusing sound excerpts creates two opposite-order windows in one recording; reversing the query exchanges which answer is correct. Training first learns this query-dependent choice, then compares interval sets by coverage, event count, and boundary accuracy. Across SpotSound-Bench, Clotho-Moment, and UnAV-100, RelTwin improves SpotSound-A and TimeAudio by up to 4.23 and 14.48 mIoU points, respectively. In matched candidate-supervision experiments, jointly correct localization of inverse-query pairs improves by 31.25 percentage points over paired supervised fine-tuning. The advantage persists after silence timing changes. RelTwin thus trains query-specific window selection while retaining direct timestamp generation at inference.

Index Terms— Audio temporal grounding, timestamp answer contrasts, query–window binding, paired supervision

1. INTRODUCTION

Audio temporal grounding takes a recording and a text query and returns the time intervals containing the requested events [1, 2]. When sounds recur, recognizing their categories does not determine which occurrence answers the query. A recording can contain both “dog then rooster” and “rooster then dog.” These queries should select different windows. Returning the same window for both is a *query–window binding error*: the model locates real sounds but selects the wrong event sequence for one query.

Existing methods improve several parts of grounding. Weakly supervised models learn from clip-level captions [3], while audio-language models support audio-conditioned text generation [4, 5, 6]. TimeAudio introduces temporal markers

and absolute time-aware encoding [7]; SpotSound interleaves timestamps with audio tokens and trains on queries for absent events [8]. CLAP aligns audio and text [9]; counterfactual caption learning [10], T-CLAP [11], and CompA [12] address distinctions between descriptions, including event order and composition. CoSTALA combines spatio-temporal contrast with local alignment and feature consistency [13]. For video grounding, SHINE uses negative queries and saliency ranking [14]. RelTwin focuses on a different supervision target: the complete timestamp answers for competing windows in the same recording.

These advances motivate a complementary question: how should a model choose between two windows that both contain the queried sounds? Sequence supervision rewards each correct timestamp answer individually, without explicitly comparing it against another window under the same query. RelTwin adds that comparison at the level of complete answers. We call these *locally valid timestamp contrasts*: each window contains a real event sequence, and the competing answer becomes correct when the query is reversed.

RelTwin (Fig. 1) constructs opposite-order windows from shared sound excerpts. Holding the recording fixed, inverse queries exchange the correct window. The model learns to favor the matching answer and to switch its preference with the query. A second training stage compares interval sets with different coverage, event counts, and boundaries. Both stages use the grounder’s own timestamp-answer scores, enabling adaptation of SpotSound-A and TimeAudio while preserving direct generation without candidate scoring or reranking at inference.

Experiments connect this training signal to localization and query-specific selection. Both backbones improve on three public benchmarks. A matched comparison isolates candidate supervision: identical answers to inverse queries fall from 37 to 1 out of 160 pairs, and jointly correct localization rises by 31.25 percentage points. The advantage persists when silence timing changes while event waveforms and order are preserved. Together, public evaluations and controlled paired tests examine standard grounding performance and the ability to select the occurrence requested by the query.

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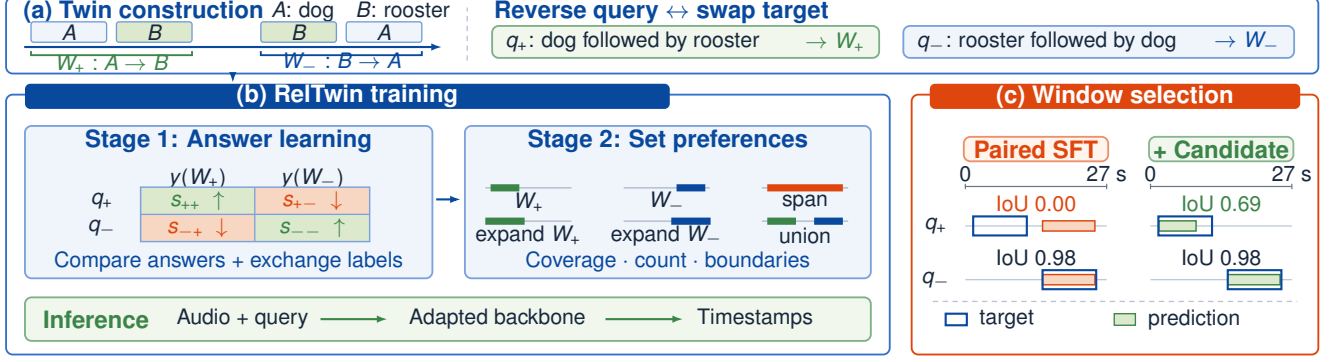


Fig. 1. RelTwin. (a) Shared excerpts form opposite-order windows $W_{\pm}$ for queries $q_{\pm}$. (b) Green/orange cells mark matching/competing timestamp answers; arrows indicate the desired score changes. Stage 2 compares interval sets. (c) SpotSound-A paired SFT versus candidate supervision: outlines are targets, filled bars are predictions, and numbers are set-IoU.

Table 1. Temporal grounding across three benchmarks with RelTwin adaptation of TimeAudio and SpotSound-A (%; higher is better). Baseline results: SpotSound [8, Table 3]. Δ : gain over the corresponding backbone. Bold marks column maxima.

Method	SpotSound-Bench			Clotho-Moment			UnAV-100 subset		
	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU
<i>Non-LLM models</i>									
WTATG	53.0	34.5	38.4	12.1	6.3	9.1	53.0	37.0	38.4
AM-DETR	25.5	15.0	19.5	89.8	88.0	80.9	59.0	46.0	42.8
<i>Proprietary models</i>									
Gemini-2.5-Flash	34.8	27.8	25.7	45.7	33.7	36.9	48.0	37.0	35.6
Gemini-2.5-Pro	20.0	17.3	19.7	40.4	32.5	32.5	39.0	35.0	34.6
<i>Open-source models and RelTwin adaptation</i>									
Kimi-Audio	3.0	1.3	3.2	0.7	0.1	0.9	8.0	2.0	5.3
Qwen2-Audio	5.5	1.8	4.4	6.1	1.3	5.7	14.0	4.0	9.7
Audio Flamingo 3	8.8	2.8	7.6	32.9	21.8	22.6	35.0	24.0	25.0
TimeAudio	7.8	1.8	10.2	39.5	24.9	28.6	21.0	7.0	16.0
TimeAudio + RelTwin	24.25	13.25	18.36	45.63	30.92	34.08	44.00	25.00	30.48
Δ vs. TimeAudio	+16.45	+11.45	+8.16	+6.13	+6.02	+5.48	+23.00	+18.00	+14.48
SpotSound-A	76.5	59.5	57.9	93.4	91.0	85.6	86.0	74.0	69.8
SpotSound-A + RelTwin	78.00	62.00	59.43	93.61	91.01	86.68	89.00	79.00	74.03
Δ vs. SpotSound-A	+1.50	+2.50	+1.53	+0.21	+0.01	+1.08	+3.00	+5.00	+4.23

2. METHOD

RelTwin has three parts (Fig. 1): twin construction creates competing windows; query-conditioned answer learning selects the matching window (Stage 1); set-level preference learning compares coverage, count, and boundaries (Stage 2). For recording x of duration T and query q , the output is an interval set $S = \{[a_k, b_k]\}_{k=1}^n$, with $0 \leq a_k < b_k \leq T$.

2.1. Twin construction

To make the query determine window selection, we arrange the same ESC-50 excerpts [15], A and B , as $A \rightarrow B$ and $B \rightarrow A$ in one recording. Targets $W_{\pm}$ span each sequence, including its gap; queries $q_{\pm}$ request the respective orders. Both

windows contain the same categories, but W_- is negative for q_+ and positive for q_- . We alternate which order appears first.

2.2. Query-conditioned answer learning

Stage 1 compares complete answers under each query. We serialize S as comma-separated spans for SpotSound-A or temporal tokens for TimeAudio. Answer tokens $y(S)$ exclude the prompt and receive the mean log-probability

$$s_{\theta}(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_{\theta}(y_t | x, q, y_{<t}). \quad (1)$$

The 2×2 matrix $M_{ij} = s_{\theta}(x, q_i, W_j)$ indexes queries by rows and answers by columns, $i, j \in \{+, -\}$ (Fig. 1b). Its diagonal entries are the correct query-answer pairs. Sequence supervi-

sion minimizes $\mathcal{L}_{\text{seq}} = -(M_{++} + M_{--})/2$. Candidate supervision additionally trains each diagonal score to exceed the competing answer in its row. For $p_i(j) = \text{softmax}_j(M_{ij}/\tau)$ with $\tau > 0$,

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_+(+) + \log p_-(-)]. \quad (2)$$

Query reversal should also reverse the answer preference. We align the two distributions after exchanging candidate labels using the Jensen–Shannon divergence $\mathcal{L}_{\text{JS}} = \text{JS}(p_+, \Pi p_-)$, where Π swaps the candidates. To retain ordinary localization supervision, we rehearse examples (x_r, q_r, S_r) with $\mathcal{R}_1 = -s_\theta(x_r, q_r, S_r)$. Stage 1 optimizes

$$\mathcal{L}_1 = \mathcal{L}_{\text{seq}} + \lambda(\mathcal{L}_{\text{cand}} + \gamma\mathcal{L}_{\text{JS}}) + \beta\mathcal{R}_1. \quad (3)$$

2.3. Set-level preference learning

Stage 2 extends selection to coverage, count, and boundaries through six candidates: $W_\pm$, their expanded versions, their enclosing span, and their union. Expansions expose boundary excess; the span includes intervening background; the union adds the competing occurrence. Their relative quality becomes the answer-level training signal.

For candidate S_j and target W_i , a four-axis quality vector h_{ij} measures set-IoU, threshold margin, count agreement, and boundary accuracy. The threshold axis averages soft F1 at $\eta = 0.3/0.5$. Each interval’s best-match IoU u contributes $\max(0, u - \eta)/(1 - \eta)$; averaging over predictions and targets gives soft precision and recall. With n_S denoting interval count, count agreement is $1 - |n_{S_j} - n_{W_i}| / \max(n_{S_j}, n_{W_i}, 1)$. Boundary accuracy is $\exp(-e)$: e averages the summed endpoint error for each target’s best-IoU match, divided by $2 \max(b - a, 0.1)$ for target $[a, b]$.

Among K candidates, let d_{ij}^+ and d_{ij}^- count those dominated by and dominating S_j . Dominance means no worse on any axis and better on at least one. We define utility as

$$v_{ij} = \frac{1}{2} \left[\frac{1}{4} \sum_{\ell=1}^4 h_{ij\ell} + \frac{d_{ij}^+ - d_{ij}^-}{K - 1} \right]. \quad (4)$$

Mean quality preserves graded differences; dominance rewards improvements without sacrificing another axis. Soft targets $t_i = \text{softmax}(v_i/\tau_q)$ assign higher probability to better interval sets. The same answer scores, normalized over all six candidates, learn this distribution through $\mathcal{L}_{\text{set}} = -\frac{1}{2} \sum_{i,j} t_i(j) \log p_i(j)$. The six-way permutation $\Pi = (2, 1, 4, 3, 5, 6)$ swaps targets and expansions, fixing the span and union.

Replay candidates perturb boundaries, add/drop events, merge spans, split the longest interval, or cover the recording. The same quality rule defines their listwise loss $\mathcal{L}_{\text{set},r}$. A frozen base-model distribution p_0 anchors replay preferences p_r . Stage 2 optimizes

$$\begin{aligned} \mathcal{L}_2 &= \mathcal{L}_{\text{seq}} + \lambda(\mathcal{L}_{\text{set}} + \gamma\mathcal{L}_{\text{JS}}) + \beta\mathcal{R}_2, \\ \mathcal{R}_2 &= \mathcal{R}_1 + \lambda\mathcal{L}_{\text{set},r} + \kappa \text{KL}(p_0 \| p_r). \end{aligned} \quad (5)$$

Both backbones share Eq. 1. Training updates LoRA parameters and, for TimeAudio, temporal/audio connectors. Inference remains direct timestamp generation from (x, q) , without twin construction or candidate scoring.

3. EXPERIMENTS

3.1. Experimental setup

Public benchmarks. We evaluate general localization on SpotSound-Bench (400 queries, 387 audios) [8], Clotho-Moment (6,649 queries), and UnAV-100 (100 queries, 77 audios) [2]. We report set-mIoU and R1 at IoU 0.3/0.5, clipping intervals to recording duration.

Twin data and relation evaluation. Forty ESC-50 classes yield 256 training recordings, 1,024 relation queries (512 pairs), and 512 category queries. Templates “followed by” and “and then” give two inverse-query pairs per recording. Relation development uses 10 held-out classes, 80 recordings, and 320 queries (160 pairs), with disjoint classes and original Freesound IDs at adaptation. It tests query-specific selection; timing intervention tests persistence under changed silence (Section 3.4).

Joint localization and relation selection. For interval unions, $\text{IoU}_{\text{set}}(S, G) = |S \cap G| / |S \cup G|$ (zero for an empty union). Given predictions $\widehat{W}_i$, let $u_{ij} = \text{IoU}_{\text{set}}(\widehat{W}_i, W_j)$. PairAcc requires $\min(u_{++}, u_{--}) \geq 0.5$. Swap error occurs if $u_{++} \leq u_{+-}$ or $u_{--} \leq u_{-+}$, including ties. JointPairAcc requires PairAcc success without Swap error. Inspired by paired compositional evaluation [16], it distinguishes query-specific selection from coverage: returning the union of two disjoint, equal-length targets passes PairAcc (IoU 0.5) but fails JointPairAcc. We report pair-level percentages; same-answer counts require identical parsed interval lists.

RelTwin training. TimeAudio starts from its official checkpoint with 384/128 updates in Stages 1/2. Its 31.76M trainable parameters use learning rates 2×10^{-6} for LoRA [17] and 5×10^{-6} for temporal/audio connectors. Category queries target both occurrences of A or B ; category- and relation-query pairs enter the paired objective at 3:1. Replay mixes Clotho-Moment training [2], synthetic scale/cardinality, TEMPO training [18], and AudioGrounding training [1] at 4:3:2:1. We use $\lambda = \gamma = 1$, $\beta = 0.75$, $\kappa = 0.2$, $\tau = 1$, $\tau_q = 0.15$, and 15% interval expansion (minimum 0.05 s).

Matched mechanism experiment. Table 2 isolates $\mathcal{L}_{\text{cand}}$ on SpotSound-A after Stage 1: $\gamma = 0$, $\beta = 0.5$, and $\lambda = 0$ (paired SFT) or 1 (candidate supervision). Both share initialization, relation/rehearsal data, data order, and 256 AdamW [19] updates at 5×10^{-6} , with zero weight decay and gradient clipping 1. LoRA uses rank 8, $\alpha = 16$, dropout 0.1, and 20.19M parameters. Each update accumulates one pair and one rehearsal query. Candidate supervision adds four gradient-free scores and weighted gradient recom-

Table 2. Mechanism experiment: isolating RelTwin’s candidate supervision on SpotSound-A. (b) Matched data and updates; (c) mean $\pm$ sample SD. Joint: JointPairAcc; same-answer: inverse-query pairs with identical timestamp answers. Scores are percentages except counts.

Method	SpotSound	Relation development (160 inverse-query pairs)			
	mIoU $\uparrow$	mIoU $\uparrow$	Joint $\uparrow$	Swap error $\downarrow$	Same-answer pairs $\downarrow$
<i>(a) Backbone reference: shared evaluation protocol</i>					
SpotSound-A	58.42	37.64	0.63	65.00	15/160
<i>(b) Matched training comparison</i>					
Paired SFT	58.42	74.58	53.13	40.00	37/160
Candidate supervision	59.43	88.55	84.38	7.50	1/160
Gain over paired SFT	+1.02	+13.97	+31.25	−32.50	−36 pairs
<i>(c) Candidate supervision: three training seeds</i>					
Candidate supervision	59.16 $\pm$ 0.26	87.05 $\pm$ 1.43	81.46 $\pm$ 2.53	8.13 $\pm$ 0.63	(1, 0, 0)/160

putation to SFT’s two positive-answer backpropagations.

Inference. Decoding is greedy, one beam, with at most 128 new tokens. The mechanism experiment uses 16 kHz mono audio and bfloat16.

Timing intervention. We recompose 80 development recordings with 1.5 s lead-in, 0.25 s within-window gaps, 3 s between windows, and 2 s tail, preserving event waveforms, order, and queries. Frozen matched models evaluate one template (160 queries). Timing changes for 40 BA-first recordings; 40 AB-first recordings remain waveform-identical.

Comparison and statistics. Table 2 evaluates the backbone and training variants under shared inference. TimeAudio uses seed 0. Mechanism point estimates use seed 0, with matched SFT; Table 2(c) separately reports seeds 0/1/2 as mean $\pm$ sample SD.

We use 20,000 audio-group bootstrap resamples for 95% CIs; relation tests resample 39 connected source-sharing groups (largest: 18 audios). The intervals quantify sampling variation for the selected checkpoints.

3.2. Grounding performance across benchmarks

RelTwin improves both backbones on all three benchmarks (Table 1). The mIoU gains range from 1.08 to 4.23 points for SpotSound-A and 5.48 to 14.48 for TimeAudio. Adapted SpotSound-A leads the listed methods. Gains on both backbones support adaptation across timestamp representations.

3.3. Timestamp contrasts recover query-specific selection

With matched data and updates, candidate supervision improves SpotSound mIoU by 1.02 points over paired SFT (95% audio-group CI: [0.23, 1.81]; Table 2).

JointPairAcc rises by 31.25 points (source-group CI [22.35, 40.50]). Identical-answer pairs fall from 37 to 1 out of 160; 27 of those 37 become jointly correct. Figure 1 illustrates this recovery. Across all pairs, transitions to joint correctness outnumber regressions by 51 to 1.

The relation mIoU gain is 13.97 points (CI [9.84, 17.51]). Three-seed mean JointPairAcc is 81.46, with same-answer counts of 1, 0, and 0. Both windows contain the queried categories, so recovering distinct, jointly correct answers demonstrates relation selection beyond category presence.

3.4. Binding gains persist after timing intervention

On the 40 recordings with altered silence timing, candidate supervision retains a 20-point JointPairAcc advantage (87.50 vs. 67.50; CI [2.94, 33.33]), compared with original scores of 85.00 and 42.50. The advantage persists when event waveforms, order, and queries are fixed but silence timing changes.

Together, public gains test transfer, paired tests isolate selection, and timing changes test position-independent relation learning; set-level learning extends this decision to coverage and boundaries.

4. CONCLUSION

RelTwin turns competing, acoustically supported answers into supervision for query-specific grounding. Inverse queries exchange which answer is correct. Gains across two backbones and three benchmarks, together with matched comparisons and timing intervention, support this answer-level learning approach while preserving direct generation.

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Compliance with Ethical Standards. We use public ESC-50 [15], SpotSound [8], Clotho-Moment, UnAV [2], TEMPO, and AudioGrounding data, including programmatically constructed sequences. The study involved no new recordings, participant recruitment, human or animal experiments, or listening studies.

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